MAPPA: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom mapping is often treated as if every reaction had a single correspondence between reactant and product atoms. In practice, the same endpoints can support multiple plausible mappings, while molecular symmetry can generate many equivalent relabelings that add complexity without adding new chemistry. We introduce MAPPA, a graph-based atom-mapping method designed to preserve meaningful alternatives without explicitly enumerating symmetry-equivalent permutations. MAPPA grows matched fragments between reactants and products, branches when distinct fragment placements remain possible, and represents symmetry-related atom assignments as compact, correlated families. A cut-sweep strategy perturbs fragment growth to recover alternatives that greedy search would otherwise discard, while a symbolic decoder translates the resulting mapping families into distinct bond-event patterns. Across the Golden benchmark, MAPPA recovers annotated correspondences more consistently than the comparison methods evaluated here while retaining alternative mappings in an inspectable representation. On reactions with coordinates and Wiberg bond orders, it also reveals bond-event alternatives that can be missed when a single mapping is selected or when mappings are chosen only by minimizing the number of bond changes. Rather than forcing a reaction into one correspondence too early, MAPPA preserves the meaningful possibilities so they can be examined and interpreted in subsequent chemical analysis.

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1 Introduction

Atom-to-atom mapping assigns corresponding atoms across a reaction. It is used to identify reaction centers, extract reaction templates, and prepare endpoint structures for mechanistic calculations. Importantly, a reaction may not have a single chemically meaningful atom correspondence: different mappings can imply different reaction centers or bond-change patterns, while molecular symmetry can produce multiple atom assignments that are chemically equivalent. For downstream analyses, it can therefore be useful to keep more than one plausible mapping rather than committing to a single correspondence too early. The challenge is to distinguish genuinely different mapping possibilities from permutations that arise only from molecular symmetry.

Existing approaches use learned correspondence signals or graph-based optimization. RXNMapper derives mappings from attention in a reaction language model (15); LocalMapper learns from chemist-labeled reactions (2); and SLAPMapper combines iterative label refinement with sequential linear assignment (9). The Golden benchmark provides a common set of annotated reactions for assessing mapping recovery (11). For applications requiring alternatives, recovering the annotation somewhere in a returned family is a different objective from producing one correct top-ranked mapping.

Weighted endpoint graphs introduce a further distinction. A permissive matching tolerance can preserve a fragment despite changes in its bond orders, but the corresponding atom shuffles need not preserve the bond events assigned at a finer threshold. Replacing a compressed family by one representative can therefore hide distinct transformations. Likewise, greedily growing one fragment can prevent a neighboring fragment from reaching an alternative assignment.

MAPPA addresses these issues by branching over structural fragment choices while keeping symmetry-related atom bijections compressed (Figure 1). Continuous growth and alternative placements determine the structural choices. Default MAPPA uses one seed ordering per cut and the cut sweep. We introduce a cut-sweep strategy that changes the search conditions to counteract greedy fragment commitment; it is part of default MAPPA, and we also apply it to SLAP to test its benefit beyond our fragment search. Its final representation groups unordered matched fragment pairs and retains the allowed mapping alternatives within each group. A constraint-based decoder projects these families onto distinct bond-event patterns. The method searches for structurally compatible alternatives without requiring the reference transformation to minimize the number of bond changes.

2 Method

Algorithm 1 connects the fragment-level operations in Figure 1. A live branch stores the fragments already placed, their atom assignments, and the compressed choices still allowed by those decisions. Growing and placing the next fragment conditions every later continuation. The matching routine is treated as an operation here; its compatibility tests and symmetry representation are defined below. For bidirectional search, the search phase is applied in each orientation, retaining the orientation of each returned family.

The forks in the search phase represent different fragment placements, not one branch for each symmetry-related bijection. The cut sweep changes the conditions of this search; a single mask can expose a mapping with several bond events. Event decoding is a separate operation on the retained output. Search caps are recorded as truncation, not treated as proof that no alternatives exist.

2.1 Weighted matching and continuous growth

Let V_R, V_P be the reactant and product atom sets, Z_R, Z_P their element labels, and W_R, W_P their symmetric bond-weight matrices. A partial mapping $m : D \subseteq V_R \to V_P$ is injective and element compatible. A mapping assigning every source atom is bijective when the endpoint element counts agree. Explicit hydrogen atoms participate in the search. Molecular coordinates can accompany the graphs but do not determine the growth decisions.

The active source graph contains edges with weight at least b. Starting from a seed atom, MAPPA grows a connected source fragment F while carrying multiple compatible product placements. A frontier queue

Algorithm 1 MAPPA: from fragment growth to alternative bond-event candidates

Require: Oriented endpoint graphs, seed orders, optional anchors, search budgets
Ensure: Compressed final branches; optionally, distinct bond-event candidates
$\mathcal{S} \leftarrow \varnothing$ {collected terminal branches}
for all conditions: uncut source graph and each single-edge source mask **do**
  for all requested seed orders under this condition **do**
    Initialize live branches with the supplied anchors
    repeat
      for all seeds in the current order **do**
        for all live branches eligible to grow from this seed **do**
          Grow a fragment to saturation, conditioned on this branch {Fig. 1a}
          if no admissible fragment placement is found **then**
            Retain this branch for later seeds
          else
            Fork one continuation per distinct structural fragment placement
            Commit each placement; retain its compressed symmetry and constraints
            Admit the continuations within the branch budget
          end if
        end for
      end for
    until all branches are finished or a full pass makes no progress
    Add terminal branches and their search-stop status to $\mathcal{S}$
  end for
end for
Group final branches by unordered fragment pairs; retain their family union
Expose compressed branches with constraints and provenance
if bond-event candidates are requested as separate post-processing **then**
  Decode distinct event classes from complete families using the original matrices
  Report witnesses, supporting families, and completeness status {Fig. 1c}
end if

prioritizes stronger source edges. To add an unassigned source atom u with product image v, a placement must satisfy

$$Z_R(u) = Z_P(v), \quad v \notin m(D), \quad W_P(v, m(r)) \geq b, \quad |W_R(u, r) - W_P(v, m(r))| \leq \tau_{\text{iso}} \tag{1}$$

for every active, nonrelaxed source edge (u, r) into F. Injectivity is enforced against all current assignments in D, including established fragments. All relevant fragment bonds are checked, rather than only the frontier edge that proposed the extension. Source nonedges do not impose negative matching constraints, so additional product bonds remain possible.

An accepted extension enlarges the fragment and filters its compatible placements. Growth continues even if only one placement remains. An unsupported edge is deferred, and other frontier edges can still be explored. When growth reaches an established fragment, ordinary continuation respects its existing assignments. A fragment saturates when no further admissible extension remains; structurally distinct surviving placements then generate separate continuations for the remaining atoms.

This is a greedy search along each trajectory: an accepted extension can remove placements that would support a different fragment boundary. MAPPA broadens coverage through cut sweeps and seed orderings.

2.2 Cut sweep to vary greedy fragment boundaries

Our cut-sweep strategy reruns growth under alternative source-edge conditions. In addition to the uncut graph $G_R = (V_R, E_R)$, it searches a graph with one eligible edge e removed, $G_R^{(e)} = (V_R, E_R \setminus \{e\})$, for each edge at or above the cut threshold. The evaluated cut threshold equals the active-graph floor $b = 0.2$, and eligible edges include bonds to H. Each subrun retains the atom set and the original endpoint weight matrices.

Removing e changes both growth and matching: the frontier cannot extend along that edge, and its bond-preservation test is omitted from Equation (1). An extension accepted in the uncut run may otherwise merge

atoms into a fragment and discard alternative placements. Under the cut condition, that growth route is unavailable; a fragment can stop at a different boundary, grow through another route, or leave atoms for a later seed. The missing preservation test also admits placements previously excluded by e. Thus the sweep counteracts greedy commitment by changing the conditions under which growth proceeds. It does not force the endpoints of e into different final fragments or require a bond-breaking event: they may reconnect through other source edges, and the final correspondence may preserve their bond.

MAPPA takes the union of the uncut and single-edge subruns before final branch deduplication. The default one-seed mode uses one seed ordering per subrun, so it already includes the full cut sweep. There are $|E_R|+1$ subruns at the evaluated thresholds. A single-edge cut describes the search perturbation, not the number of bond events in a returned mapping. One removed constraint can redirect subsequent fragment growth and placements throughout the reaction, exposing an alternative with several bond formations, cleavages, or bond-order changes. MAPPA therefore need not enumerate combinations of final bond events to discover multi-event alternatives. Bond-event decoding uses the original matrices, so a search cut itself contributes no event or event-count penalty. Growth remains greedy within each conditioned subrun; the union broadens coverage without making the search exhaustive.

2.3 Automorphisms and correlated mapping families

The distinction between structural branches and explicit bijections is central to the design. Even one fragment family can contain combinatorially many assignments: k independently exchangeable methyl-hydrogen triplets admit $(3!)^k$ bijections. Expanding those automorphisms into separate mappings would multiply the work without adding fragment choices. MAPPA instead stores the permitted permutation actions and their constraints, and decodes event alternatives directly from this representation. Branch caps bound retained alternatives within each growth call; they do not guarantee a small global search. This avoids explicitly listing symmetry-related bijections within each retained family.

Compatible placements are stored using a representative assignment, unresolved assignment blocks, and permutation generators. During growth, graph canonicalization accounts for the current fragment, locked assignments, and boundary information. Each saved transition records its action generators together with conditioning and fragment-preservation constraints; their conjunction defines the allowed mapping relation. Canonical labeling and automorphism computation use nauty (12).

Matching tolerance and automorphism equivalence have different roles. The relation $|x-y| \le \tau_{\text{iso}}$ is generally not transitive, so it does not by itself define a permutation group. To describe an intrinsic symmetry shared by a final fragment pair, we instead color product bonds by their responses to the fragment's matching tests. For a product weight w and the set $\mathcal{W}$ of active reactant weights for the same element pair, the label records

$$\chi_{\mathcal{W}}(w) = \left(\mathbf{1}[w \ge b], (\mathbf{1}[w \ge b \ \wedge \ |w-v| \le \tau_{\text{iso}}])_{v\in\mathcal{W}}\right). \tag{2}$$

An exact automorphism of this colored graph preserves every encoded matching test. This definition depends on the final fragment pair, not its growth order.

The saved mapping relation also contains the constraints imposed by its search decisions. We represent a flat family $\mathcal{F}_j$ by a seed mapping m_j, a sequence of allowed action groups $G_{j,1},\dots,G_{j,k}$, and final preservation constraints C_j:

$$\mathcal{F}_j = \{a_k \circ \cdots \circ a_1 \circ m_j : a_i \in G_{j,i},\ C_j(a_k \circ \cdots \circ a_1 \circ m_j)\}. \tag{3}$$

These actions encode correlated moves; atoms in a common orbit are not independently permutable. Disjoint actions commute and can be placed in a canonical order. The order of overlapping actions is retained. A sequence of groups is not generally equivalent to the group generated by their union.

The intrinsic automorphism of a fragment pair is shared metadata, not permission to enlarge its saved family. Multiple admissible seed mappings can occupy different intrinsic symmetry orbits, and different search constraints can restrict the allowed actions. MAPPA therefore retains these alternatives explicitly inside the compressed representation.

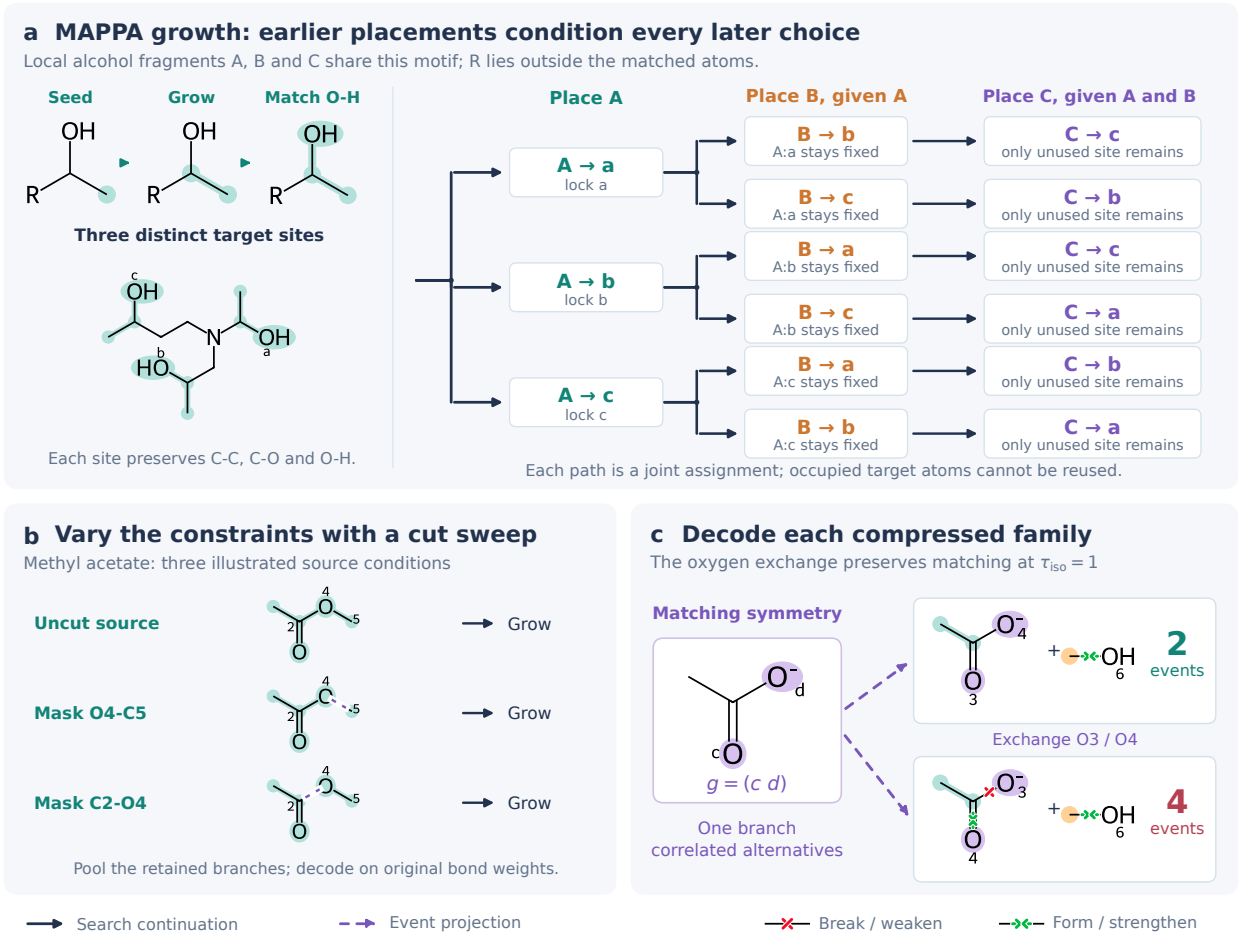


Figure 1 MAPPA: conditional fragment branching and compressed symmetry. (a) A local alcohol motif grows while retaining its O–H bond. Three sites in the depicted target have distinct molecular environments. For three source fragments sharing this motif, placing A fixes one site; B then branches over the two unused sites, and C takes the remaining site. The tree illustrates local conditioning and injectivity; R denotes atoms outside the matched subgraph, and this is not a complete reaction-search trajectory. All attached hydrogens in the illustrated local matches are checked. (b) The cut sweep searches the uncut source graph and graphs with one source edge masked. The illustrated ester masks change the growth and preservation constraints; their outputs are pooled. Masking an edge does not itself assign a bond event. (c) Exchanging the two acetate oxygen images preserves the fragment's element and bond-weight matching tests at $\tau_{\text{iso}} = 1$, but changes two heavy-atom events into four at threshold 0.5. Neither acetate oxygen carries H. Solid black lines retain the endpoint bonds. Red × marks on the bonds indicate cleavage or weakening; paired green arrows pointing inward indicate formation or strengthening. Pastel colors track fragment membership or symmetry sites. Dashed arrows are event projections within a family, not additional search branches. Saved constraints govern allowed symmetry actions. These formal-bond-order event signatures illustrate MAPPA decoding, not distinct mechanisms or resonance-normalized transformations.

2.4 Unordered final branches

During search, history and boundary conditions matter because they determine valid continuation. At completion, the branch identity is instead the unordered collection of matched atom sets,

$$\mathcal{B} = \{(R_1, P_1), \ldots, (R_q, P_q)\}, \qquad P_i = m(R_i), \tag{4}$$

under a fixed matching policy. Sorting atoms within each set and then sorting the fragment pairs removes dependence on growth order and fragment labels. Growing A before B or B before A yields one branch when the final paired sets agree. The colors within each candidate in Figure 2 illustrate these matched fragment pairs.

Each branch stores the union of its flat families. Identical normalized family records are shared, while distinct constraints or action programs remain available. This transformation preserves the represented mapping union: it changes its grouping, not its admissible assignments. Provenance links retain the discovering paths, but the original search hierarchy is unnecessary for decoding. This conservative deduplication does not assert that the remaining families are mutually disjoint.

2.5 Decoding distinct bond-event patterns

For each complete mapping, define the signed change for an unordered source pair $\{u, v\}$ by

$$\Delta_{uv}(m) = W_P(m(u), m(v)) - W_R(u, v), \qquad s_{uv}(m) = \begin{cases} +1, & \Delta_{uv}(m) \geq \delta_{uv}, \\ -1, & \Delta_{uv}(m) \leq -\delta_{uv}, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Positive events denote bond formation or strengthening; negative events denote cleavage or weakening. The event count is $E(m) = \sum_{u<v} \mathbf{1}[s_{uv}(m) \neq 0]$. The evaluation includes hydrogen atoms and all atom pairs. For the WBO collection, $\delta_{uv} = 0.3$ for pairs involving a metal and 0.5 otherwise. These thresholds are independent of τ_{iso}. Figure 2 provides a concrete example: changing the atom correspondence changes which pairs gain or lose bond order, even when the total event counts agree (A and C).

The decoder returns one bijection witness per distinct signed event class, identifying patterns related by exact source permutations that preserve all thresholded bond-change tests. Matching-preserving shuffles can therefore yield different event classes. The decoder first certifies families whose assignments all yield equivalent events and excludes families outside the requested event window. It then encodes the remaining family constraints symbolically in Z3 (5). Each solution supplies a witness and its event pattern; excluding that entire event class allows the next distinct pattern to be found. This avoids enumerating individual bijections or permutation groups.

For a window $E \leq K$, decoding is complete when every retained family is certified exhausted or excluded. A timeout leaves completeness unresolved. This certificate concerns the saved search output within the window; it does not establish exhaustive exploration of all chemically possible mappings. The default deduplication preserves families that contain different event counts, as illustrated in the heavy-atom projection in Figure 1c.

2.6 Alternative correspondences and chemical interpretation

An atom mapping summarizes the net correspondence resulting from a chemical process without specifying its intermediates or elementary steps. When that pathway is unknown, endpoint structures alone may not determine a unique correspondence. MAPPA therefore retains alternative bond-event patterns and the correlated shuffles within each, exposing one witness per event class together with its supporting families and conditional symmetry queries. Distinct patterns can have equal event counts, and minimizing the count need not recover the reference correspondence.

Figure 2 illustrates both points with Golden case 9. MAPPA recovers the annotated correspondence, in which the alcohol oxygen enters the product ring and the ketone oxygen leaves in water (A). Another candidate exchanges these oxygen origins (B); a third also exchanges two carbon assignments (C). Candidate B has fewer total events than the reference, while A and C have equal totals but different correspondences. Thus, selecting either one bijection or only the lowest event count discards information about the alternatives. These are selected witnesses from the search, not an exhaustive decoded set or validated reaction pathways.

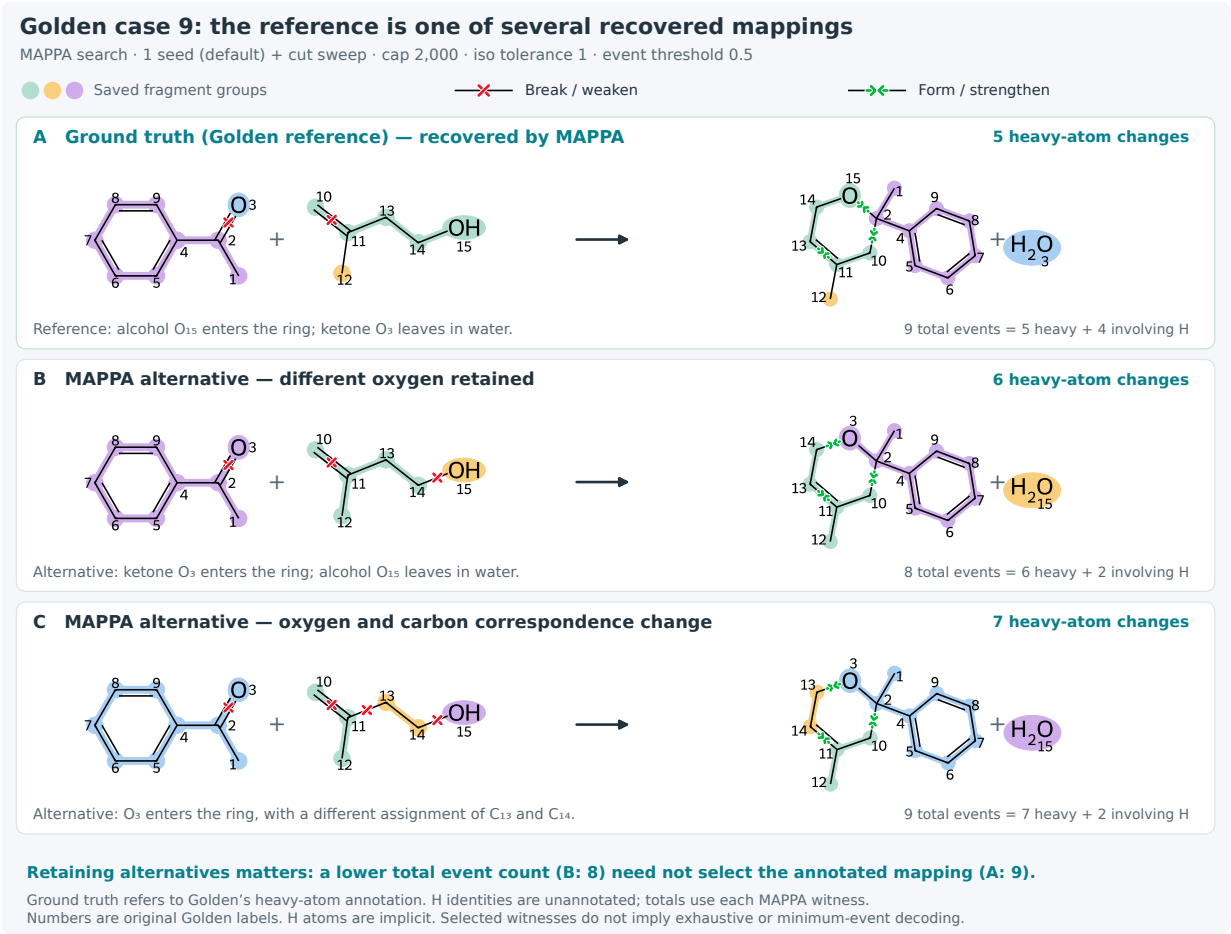


Figure 2 A recovered reference and alternative correspondences from Golden case 9. A matches all 15 annotated heavy-atom pairs. B exchanges the oxygen origins; C also exchanges C_{13}/C_{14}. Pastel colors identify matched fragments. Black lines show endpoint bonds; red × marks indicate cleavage or weakening, and green inward arrows indicate formation or strengthening. H atoms are implicit in the drawing; event totals include H. Each witness minimizes H-involving changes for its fixed heavy-atom correspondence. "Ground truth" denotes the dataset annotation, not a verified pathway. The three witnesses come from one-seed MAPPA sweep search at cap 2,000 and $\tau_{\text{iso}} = 1$, without reference labels; events use threshold 0.5.

3 Evaluation

3.1 Datasets and comparison protocol

We evaluate reference recovery on all 1,851 Golden reactions (11) and bond-event alternatives on 140 coordinate/WBO endpoint pairs. Golden inputs use formal bond orders and explicit H, with reference labels removed before search. Recovery requires the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical symmetries preserving charge, isotope, H count, and stereochemistry. Failures and unresolved cases remain in the denominator. This measures reference inclusion among returned alternatives, not their ranking.

LocalMapper provides the prior accuracy-SOTA baseline (2). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration. SLAP reported 86.9% on that dataset version, counting reference inclusion among its candidates (9). These literature scores are separate from our strict re-evaluation. We additionally apply the cut sweep introduced here to SLAP; "SLAP sweep" denotes this extension, not a

Table 1 Evaluation configurations. Both use $b = 0.2$, $\tau_{\text{iso}} = 1$, explicit H, and uncut plus single-edge searches. Default MAPPA uses one seed per cut, the sweep. Golden measures the sweep-search stage; no-sweep controls retain only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Bidirectional	Reactant to product
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100 (default)	100 (default), 2,000
Evaluation	Reference-family recovery	Signed event-class coverage

published SLAP score. It is our strongest evaluated prior-method baseline for alternative recovery. Released RXNMapper, Chython, Indigo, and RDT provide further comparisons (15; 3; 6; 14).

The coordinate collection contains 18–137 explicit atoms per endpoint, including organometallic and multi-component structures. It was inspected during development and has no verified atom mappings: it is not a mapping-accuracy benchmark. We compare the lowest event count found and the patterns achieving that count. MAPPA uses supplied WBO matrices; SLAP uses XYZ-derived binary graphs. Returned SLAP heavy-atom assignments receive hydrogen refinement and are then scored on the same WBO matrices as MAPPA using Equation (5). The comparison covers those witnesses, not all possible SLAP hydrogen assignments.

Table 1 lists the evaluated settings. Search does not use reference or comparator correspondences. Full configurations, implementation versions, and reproduction scripts are provided in the repository (Appendix A).

3.2 Golden reference recovery and search cost

Default MAPPA recovers 99.08% of Golden references, compared with 97.03% for SLAP sweep (Table 2 and figure 3). The paired comparison includes 43 references recovered only by MAPPA and 5 only by SLAP. LocalMapper is the strongest evaluated method returning one bijection. Additional MAPPA seeds yield modest recovery gains at increasing search cost; two seeds do not improve recovery over the default.

The cut sweep is central to this result: it adds 345 reference recoveries for MAPPA and 135 for SLAP. Without the sweep, one-seed MAPPA has lower coverage than bidirectional SLAP. This supports varying the growth constraints to expose alternatives excluded by greedy fragment extension (Section 2.2); symmetry expansion alone cannot restore discarded placements.

Smaller-first grows fragments from the reaction side with fewer explicit atoms (including H) and matches into the other side; larger-first reverses these roles. Bidirectional search combines both outputs. It adds 31 recoveries over smaller-first alone, demonstrating that the two orientations retain complementary alternatives (Table 3).

Default MAPPA has a lower mean and median search cost than SLAP sweep, but a higher 95th percentile. Two seeds approximately double the mean cost without improving recovery here. Golden timings cover search, excluding reference verification and separate bond-event decoding. The two timing tables use different experiments and should be compared within their stated CPU and reaction sets.

Increasing the branch cap from 100 to 2,000 yields no additional swept recoveries among cases resolved at both caps, while increasing cost and unresolved outcomes at higher seed counts (Table A3). The uncut search gains 19 confirmed recoveries. Thus, a larger branch budget does not substitute for changing the growth conditions.

The default-comparator rows in Table 2 use released interfaces without prediction repair or reference-guided reranking. Binary SLAP is retained as the stronger default configuration. The other methods return at most one explicit bijection per reaction, whereas SLAP returns multiple candidates. Invalid endpoint chemistry and duplicate atom labels count as failures; the repository records their frequencies. These results describe the evaluated interfaces and scoring rule, including its difference from RDT's chemical-change criterion.

Table 2 Golden reference recovery and runtime. Recovery uses all 1,851 primary-evaluation records. MAPPA retains compressed families; SLAP returns explicit candidates; other methods return one bijection over their matched atoms. MAPPA uses cap 100 and bidirectional search. Bidirectional SLAP combines both directions and both binary/weighted modes; the default SLAP row uses binary only. Times are mean CPU seconds per reaction, excluding separate bond-event decoding. Unmarked times use the same 1,403 completed reactions on an AMD EPYC 9J14. *a*: archived completed calls (1,718–1,851 per method), CPU model unrecorded; these times are not directly comparable with unmarked times. †: LocalMapper, prior accuracy SOTA (2024). ‡: SLAP with our sweep, strongest evaluated prior-method recovery baseline.

Method	Search setting	Output	Reference recovered	Mean CPU s
MAPPA	1 seed, sweep (default)	Families	1,834 (99.08%)	0.963
MAPPA	2 seeds, sweep	Families	1,834 (99.08%)	1.619
MAPPA	3 seeds, sweep	Families	1,837 (99.24%)	2.265
MAPPA	10 seeds, sweep	Families	1,840 (99.41%)	6.993
MAPPA	1 seed, no sweep	Families	1,489 (80.44%)	0.040
SLAP, bidirectional	No sweep	Candidates	1,661 (89.74%)	0.105
SLAP, bidirectional‡	Sweep	Candidates	1,796 (97.03%)	2.793
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)	0.075[a]
RXNMapper	Default, no sweep	One bijection	1,547 (83.58%)	0.056[a]
LocalMapper†	Default, no sweep	One bijection	1,605 (86.71%)	0.294[a]
Chython	Default, no sweep	One bijection	1,561 (84.33%)	0.612[a]
Indigo	Default, no sweep	One bijection	692 (37.39%)	0.184[a]
RDT	Default, no sweep	One bijection	1,030 (55.65%)	1.021[a]

Table 3 Recovery and runtime by search orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; CPU statistics are seconds per reaction over the same 1,821 completed reactions (Apple M4 Pro CPU). Smaller-first grows from the side with fewer explicit atoms, including H; larger-first reverses the source and target. Equal-size ties use stored order and its reverse. Bidirectional output combines both searches, and its CPU cost is their sum. SLAP combines binary and weighted modes. CPU/SLAP compares means within each orientation. Timing scopes are specified in Appendix A.

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
MAPPA, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
MAPPA, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

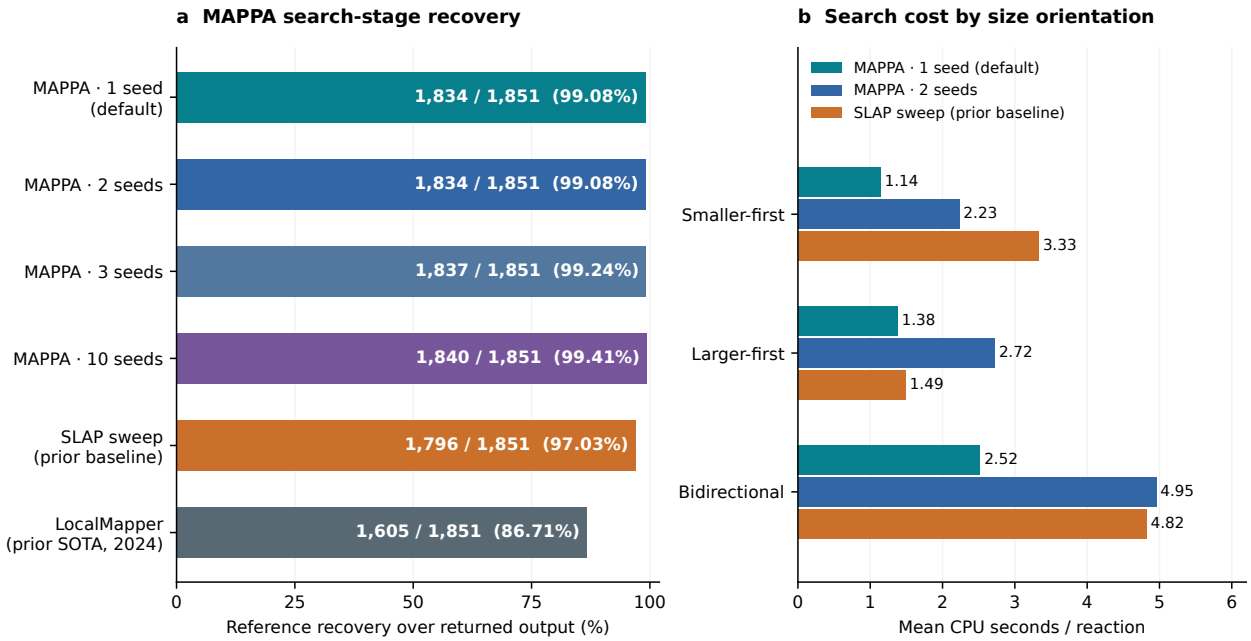


Figure 3 Golden reference recovery and search cost. (a) Reference recovery over all 1,851 reactions. MAPPA and SLAP use our cut sweep; LocalMapper is the prior accuracy-SOTA baseline and returns one bijection. All scores use our evaluator. (b) Mean search CPU over the same 1,821 completed reactions, by orientation. Bond-event decoding is excluded. Full comparisons and timing scopes are in Tables 2 and 3.

3.3 Minimum bond-event comparison on 140 coordinate reactions

We use one-seed MAPPA sweep search with a branch cap of 2,000, meaning that the search can retain up to 2,000 concurrent branches before truncating the search. This configuration returns complete mappings for all 140 reactions. With the library-default branch cap of 100, two reactions (cases 123 and 125) remain incomplete. We therefore use the 2,000-branch configuration for this comparison, while retaining 100 as the library default.

MAPPA matches or lowers the minimum event count found by both native SLAP and SLAP sweep in every reaction (Table 4 and figure 4). This does not imply recovery of every alternative: at equal minima, six SLAP-sweep patterns remain absent across four reactions. An equal count can describe different changed bonds, and a lower count alone does not establish a more plausible transformation.

Table 4 Minimum bond-event comparison on 140 reactions. MAPPA uses one-seed sweep search at cap 2,000. The first three columns count reactions by MAPPA's minimum event count relative to the comparator. The final column gives comparator minimum-event patterns recovered by MAPPA among reactions with equal minima. Counts refer to returned candidates, not proven global optima.

Compared with	MAPPA fewer	Equal	MAPPA more	Patterns at equal minima
SLAP sweep	4	136	0	158/164
Native SLAP	15	125	0	138/140

To compare alternatives above MAPPA's own minima, we decode through one event above the higher of the two SLAP minima for each reaction. MAPPA retains 162 of 168 SLAP-sweep minimum-event patterns and 151 of 160 native-SLAP patterns within these windows. The output contains 300 distinct patterns, including 166 at MAPPA's own minima. Every saved family is decoded or certified within its window; this establishes completeness for the retained families, not exhaustive structural search.

Table 5 Stage CPU time per reaction on the 140 coordinate reactions. Measurements use an Apple M4 Pro CPU. MAPPA uses one-seed sweep search at cap 2,000. These stage costs exclude MAPPA decoding and do not establish an end-to-end speed ranking. SLAP refinement covers its returned witnesses. Timing boundaries are specified in Appendix A.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.047	0.314	3.481
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

The stage timings in Table 5 cover different outputs: MAPPA search retains families for subsequent decoding, while SLAP refinement operates on returned witnesses. They therefore do not establish an end-to-end speed ranking.

A fresh run of the final implementation completed search and event decoding for 140/140 coordinate reactions. On this full set, the full pipeline takes a mean 6.07 CPU seconds per reaction (median 0.81; 95th percentile 25.47). Decoding accounts for a mean 4.65 CPU seconds (median 0.29). These measurements include search, final-family construction, decoding, and candidate output; they are separate from the search-only comparison above (Table A4).

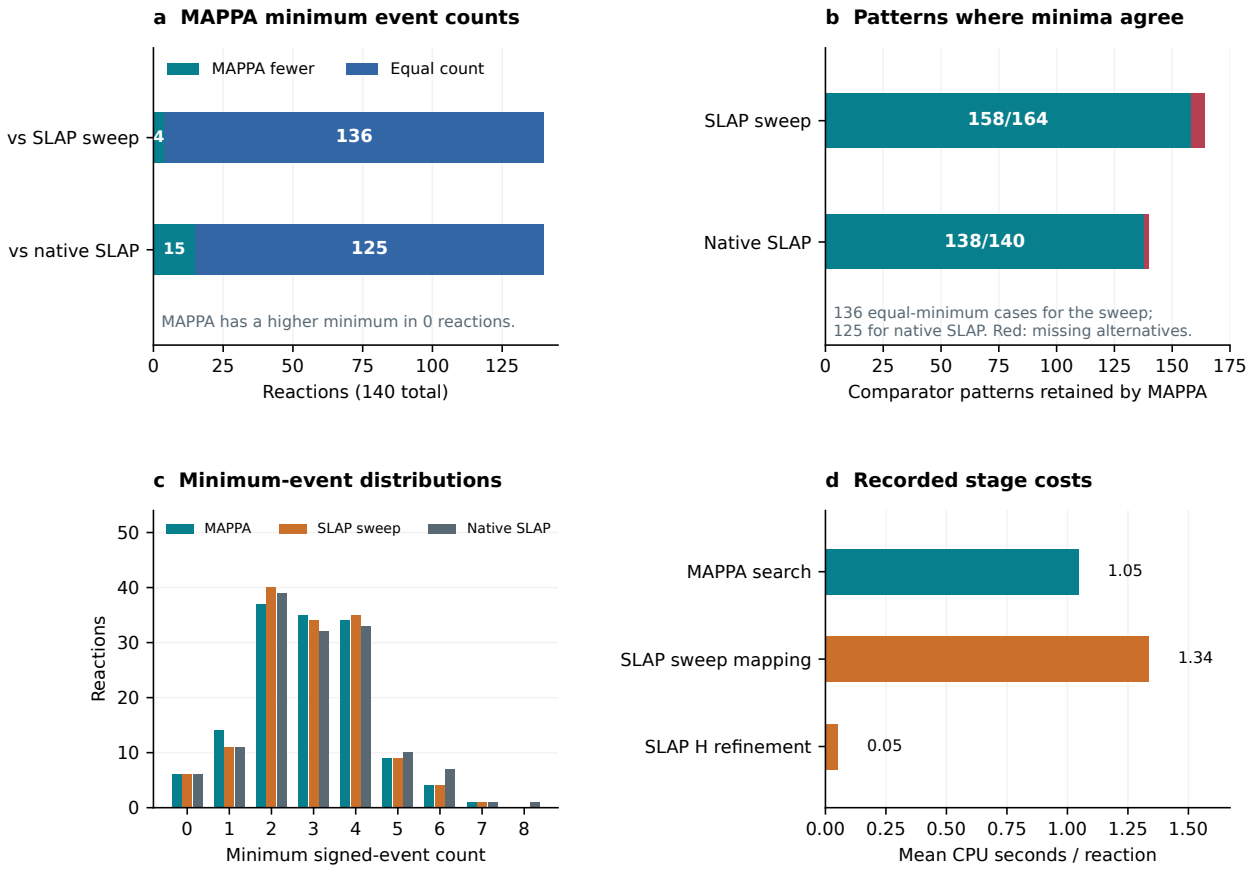


Figure 4 Bond-event comparison on 140 coordinate reactions. MAPPA uses one seed, the sweep, and cap 2,000. (a) Reactions with fewer or equal MAPPA events than each SLAP baseline. (b) Comparator minimum-event patterns recovered by MAPPA at equal minima. (c) Minimum-event count distributions. (d) Mean CPU for search and SLAP refinement, excluding MAPPA decoding (Table 5). Atom mappings are unverified; this compares bond events, not mapping accuracy.

4 Discussion

4.1 Applications and future directions

Atom correspondence is a preprocessing decision in reaction-path interpolation: the assignment determines which product position each reactant atom approaches. Atomic-index assignment is a recognized obstacle in reaction-pathway search (4). MAPPA's complete correspondences and allowed joint shuffles could be used to reorder and align endpoints before constructing initial paths for nudged elastic band (NEB) calculations (8). Different event patterns provide alternative reaction cores, while symmetry-related assignments within one pattern can give different interpolations. This offers a systematic way to prepare several TS-search starting points. Mapping the same core into a TS guess also allows its vibrational modes to be assessed against the intended bond changes.

Structure verification is another application. A generative model may return a molecule with reordered atoms or a different conformation, making direct coordinate comparison unsuitable for checking its connectivity. AAM supplies an explicit correspondence under which the target and generated structures can be compared for missing or extra connections. A complete witness that passes this comparison certifies agreement under the chosen graph representation and tolerances. Connectivity alone does not establish stereochemical identity or energetic stability, and failure to find a witness in a bounded search is not proof of a mismatch.

Such checks could also supply a verification mechanism for large language model (LLM) post-training, including reinforcement learning with verifiable rewards (RLVR) (10). For tasks that ask a model to generate a target structure or satisfy specified structural constraints, an independently checked atom correspondence could provide a task-specific reward. The reward would certify the stated structural conditions, with unresolved searches kept distinct from verified failures. This is a proposed use of the verifier; we have not evaluated an RLVR training loop or its effect on molecular generation.

For pathway and retrosynthesis analysis, alternative event patterns expose different atom origins, reaction centers, and possible fragment disconnections. These could help organize mechanistic hypotheses or compare proposed precursors while retaining symmetry-related choices for subsequent geometric analysis. One net correspondence can admit several intermediate sequences, however, and a compatible mapping need not describe a feasible reaction. Energies, reaction conditions, stereochemistry, and experimental evidence remain necessary to assess the proposed routes. The present evaluation concerns correspondence coverage and bond-event alternatives; it does not measure improvements in NEB convergence, TS-search success, or synthesis planning.

Alternative enumeration is not unique to MAPPA or SLAP; for example, an Ising-computing approach enumerates optimal mappings and reduces them by molecular symmetry (1). MAPPA's contribution is to retain conditional fragment families and expose their distinct event patterns and allowed shuffles without expanding all bijections.

4.2 Coverage and remaining search limitations

An audit of all eleven ten-order Golden misses separates unmatched-atom decisions from fragment-placement failures (Table A2). In four reactions, every retained family maps exactly one more heavy-atom pair than the reference. Such a family cannot reproduce the reference's unmatched-atom pattern through symmetry shuffling: permutations preserve the number of assigned pairs. The remaining seven lack the required correspondence within their saved families. Re-ranking or expanding those families into explicit bijections cannot supply a missing structural alternative.

Two traced examples show how greedy extension excludes a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Ordinary growth preserves bonds that these correspondences require separating. Uncapped trajectories still miss both references, while reference-constrained fragment choices establish their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposes the reference to ordinary matching; the single-edge sweep does not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries. For the other five correspondence misses, the audit does not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. Independent evaluation on annotated coordinate reactions is needed to assess generalization beyond the development collection.

Code and data

Code and reproduction instructions are available at MAPPA source repository, branch `main`. The repository guide `manuscript/REPRODUCIBILITY.md` links full benchmark configurations, software versions, per-case results, and figure-generation scripts. Golden is distributed by its original authors (11).

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A Evaluation details

A.1 Reproducibility and comparison scope

The repository's `manuscript/REPRODUCIBILITY.md` links the benchmark scripts, pinned software and input versions, full configurations, and per-case records. Table 1 gives the settings needed to interpret the comparisons. The notebook `docs/AAM_SIMPLE.ipynb` demonstrates matching, unique event candidates, and conditional symmetry queries through the Python API.

For the coordinate comparison, MAPPA uses supplied WBO matrices and SLAP uses the original XYZ-derived binary graphs. SLAP's returned heavy-atom assignments receive hydrogen refinement under its original bond-change objective, then are rescored on the WBO matrices. Thus, the comparison uses a common event score but different input representations. It covers the returned SLAP witnesses, not an exhaustive expansion of its hydrogen label families. The WBO-generation method is not recorded in the available metadata, so regeneration of these matrices from molecular structures is not established.

The paired search timings use an Apple M4 Pro CPU. Averages use 1,821 common completed reactions. Reference verification and interrupted work are excluded. MAPPA includes checkpoint I/O; SLAP excludes I/O. Higher-seed runs measured on different CPUs are excluded from this paired comparison. For the coordinate stage comparison, MAPPA search includes checkpoint I/O; SLAP mapping excludes input preparation and export, and its hydrogen refinement includes I/O. Shared event rescoring and failed or replaced attempts are excluded. Golden MAPPA timing includes symmetry finalization; SLAP includes graph construction, mapping, and export. In the primary Golden evaluation, uncut and swept SLAP have one and three unresolved cases, respectively; all MAPPA verdicts are resolved.

A.2 Default comparator protocol

Released comparators receive the same label-free Golden inputs, without prediction repair or reference-guided ranking. Invalid outputs and failures remain in the denominator. SLAP is evaluated over all returned candidates, using binary mode for the default comparison and both modes for bidirectional comparisons. The repository records the released models, versions, and adapter settings.

A.3 Decoding scope

For each coordinate reaction, the decoding limit is one event above the larger of the native-SLAP and SLAP-sweep minima. All 120,052 retained families are exhausted within these limits, without a class-count cap. Higher-event patterns are outside this completeness claim. An event class records the signed atom-pair changes, modulo source permutations preserving every thresholded bond-change test against the opposite endpoint. Each class has a concrete mapping witness.

Table A1 Complete final decoding on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

A.4 Preservation of mapping families under flattening

Flattening retains each terminal representative, its required fragment constraints, and its allowed action sequence after applying prefix locks. Disjoint actions commute. Adjacent nested groups satisfy $HG = GH = G$ when $H \leq G$, with containment checked by exact generator membership. Other overlapping actions retain their order. Identical records are shared; branches with the same unordered fragment pairs retain the union of their family records. The admissible mapping union is therefore unchanged. A shared fragment partition or automorphism group alone does not establish family equality.

The decoder certifies a family through event invariance, a valid lower bound excluding the requested window, or unsatisfiability after excluding all discovered event classes. An exhausted budget remains incomplete. This separates lossless representation, complete decoding within a window, and exhaustive structural search.

A.5 Golden nonrecoveries

Table A2 The eleven Golden nonrecoveries. Case indices are zero-based. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

Independent reconstruction agrees with the supplied references in all eleven cases. Four require a different number of matched pairs; seven require correspondences absent from the retained families. All have at least one capped direction, but this does not identify the cap as the cause. The uncapped traces discussed in the main text demonstrate greedy-growth exclusions in two cases. Reference-directed feasibility checks are diagnostic and do not contribute to benchmark recovery. Per-case certificates and traces are linked in the repository.

A.6 Branch-cap and CPU-time ablation on Golden

We reran the twelve configurations in Table A3 on all 1,851 Golden reactions using an AMD EPYC 9J14 CPU. Both search directions are unioned. SLAP unions its binary and weighted modes. Search and reference verification have separate fixed resource budgets. The recorded cap-100 recovery counts reproduce the main evaluation. At cap 2,000, unresolved counts increase with seed count. Among reactions resolved at both caps, no swept configuration gains or loses a reference recovery.

The ten-seed cap-2,000 run leaves five references recovered at cap 100 unresolved, accounting for its lower confirmed count. The bounded SLAP sweep confirms 1,794 references here, compared with 1,796 in the main evaluation: two previously recovered cases are unresolved under this study's limits. These outcomes are reported separately from verified exclusions. The paired timing cohort excludes 448 reactions because at least one configuration does not finish; the all-attempt cost column exposes the additional computational cost on those difficult reactions. Source pins, individual outcomes, directional costs, and reproduction scripts are provided with the code.

Table A3 Golden branch-cap ablation with paired SLAP baselines. Recovery counts confirmed reference inclusion over all 1,851 records; unknown cases remain in the denominator. CPU values are seconds per reaction. Paired CPU uses the same 1,403 reactions whose searches finish under every configuration. Spent CPU averages recorded search cost over all 1,851 selected attempts, including interrupted calls; it is not completed-search latency. Both timings exclude reference verification and initial imports. Superseded attempts are excluded. Cap 100 with a one-seed sweep is the default.

Method	Cap	Seeds	Recovered (%)	Unknown	Paired CPU	Spent CPU
MAPPA uncut	100	1	1,489 (80.44)	0	0.040	0.085
MAPPA sweep (default)	100	1	1,834 (99.08)	0	0.963	3.873
MAPPA sweep	100	2	1,834 (99.08)	0	1.619	7.260
MAPPA sweep	100	3	1,837 (99.24)	0	2.265	10.556
MAPPA sweep	100	10	1,840 (99.41)	0	6.993	33.211
MAPPA uncut	2,000	1	1,508 (81.47)	5	0.086	0.517
MAPPA sweep	2,000	1	1,834 (99.08)	2	4.317	30.055
MAPPA sweep	2,000	2	1,834 (99.08)	4	8.283	48.549
MAPPA sweep	2,000	3	1,837 (99.24)	6	12.201	61.890
MAPPA sweep	2,000	10	1,835 (99.14)	12	43.843	113.777
SLAP uncut	–	–	1,661 (89.74)	1	0.105	0.659
SLAP sweep	–	–	1,794 (96.92)	5	2.793	9.895

A.7 Fresh end-to-end timing on coordinate reactions

We measured one-seed, cap-2,000 cut-sweep search followed by the public event decoder on the Apple M4 Pro CPU. CPU times are reported per reaction. The event thresholds and windows match the bond-event comparison, with no class-count cap. All 140 completed reactions reproduce the archived event-class sets and family counts. Median end-to-end wall time is 0.96 seconds. Timing includes input loading, checkpoint I/O, final-family construction, decoding, and candidate serialization; initial imports and reference-set verification are excluded. All 140 reactions finish within the five-minute watchdog. Detailed timing records and reproduction instructions are provided with the code.

Table A4 MAPPA search plus bond–event decoding. CPU seconds per completed reaction in the fresh 140-reaction pass. The final row measures the whole pipeline; medians and percentiles of individual stages are not additive. Completion coverage and timing scope are stated above.

Stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.257	0.300	3.468
Final-family catalogue	0.156	0.019	0.595
Bond-event decoding	4.647	0.288	24.361
Candidate serialization	0.005	0.001	0.015
End to end	6.073	0.812	25.470